

Agentic AI Embedded Skill Workshop

MATLAB Desktop Setup Instruction

Platform: Windows or macOS | MATLAB R2026a | Participant pre-work

1. Required MATLAB Version

Use the MATLAB release provided with the workshop **MathWorks trial license**.

- **MATLAB R2026a** (R2026a is the release used to prepare and test the workshop files)

★ Note

Please complete this setup **at least one day before** the workshop. Installation during the session is a recovery option only and may be blocked by your company IT policy.

If you need help, contact your Ascendas workshop coordinator before the event.

2. MATLAB R2026a System Requirements

Check that your laptop meets the MathWorks system requirements for the release:

- Windows system requirements: <https://www.mathworks.com/support/requirements/matlab-system-requirements.html>
- macOS R2026a requirements: <https://www.mathworks.com/content/dam/mathworks/mathworks-dot-com/support/sysreq/files/system-requirements-release-2026a-macintosh.pdf>

Bring a supported Windows laptop or Apple-silicon Mac, its charger, and permission to install software. MATLAB R2026a supports Apple M-series processors and is not available for Intel-based Macs. Supported macOS releases for R2026a are macOS Tahoe 26, Sequoia 15, and Sonoma 14.

3. Required Toolboxes

Install the following MathWorks products for the hands-on workshop:

- MATLAB
- Simulink
- Deep Learning Toolbox
- Statistics and Machine Learning Toolbox
- MATLAB Coder
- Simulink Coder
- Embedded Coder
- Fixed-Point Designer
- MATLAB Report Generator

4. Required Add-Ons and Support Packages

Install these add-ons from the MATLAB Add-On Explorer after MATLAB is installed:

- Deep Learning Toolbox Converter for PyTorch Models
- Deep Learning Toolbox Model Compression Library
- MATLAB Coder Interface for Deep Learning
- Windows compiler option: MATLAB Support for MinGW-w64 C/C++/Fortran Compiler
- macOS compiler option: Xcode 16 or Xcode 26 from Apple. Xcode is not installed through MATLAB Add-On Explorer.

A MATLAB-supported C/C++ compiler is required for local desktop MEX verification. Windows can use MinGW-w64 or a supported Microsoft compiler. macOS uses Xcode 16 or Xcode 26. Without a compiler, participants can complete Parts 1 and 2 and use the prepared MEX evidence in Part 3.

5. Step-by-Step Installation Instructions

5.1 Verify the Trial License Account

During issuance of the trial license, a MathWorks account is automatically created for users who do not already have one.

1. Open the e-mail sent by MathWorks.
2. Click **Start your trial now** to verify your account.

Thank you for your interest in MathWorks products. Your license expires on 2024-Oct-31. You can view your details in the [License Center](#).

If you do not have a MathWorks Account, one has been created for you. Check your email and complete the setup process.

Start your trial now

If you have trouble downloading, installing, or activating your software, [contact support](#) for help.

Sincerely,
MathWorks Sales Team

Start your trial now link

5.2 Sign In to MathWorks

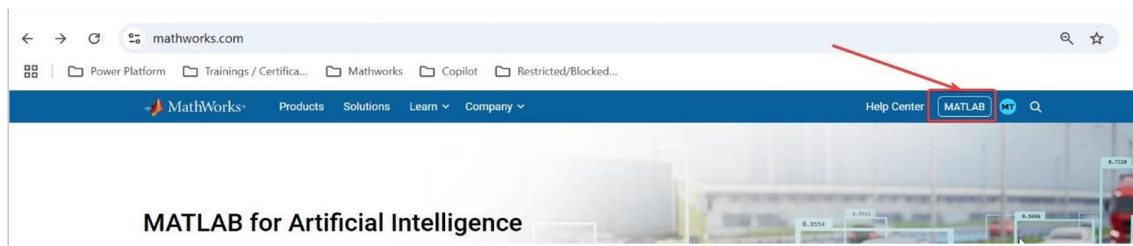
1. After verifying your account, go to <https://www.mathworks.com>.
2. Log in with your MathWorks account.
3. If you forgot your password, click **Forgot Password?** and follow the reset instructions.



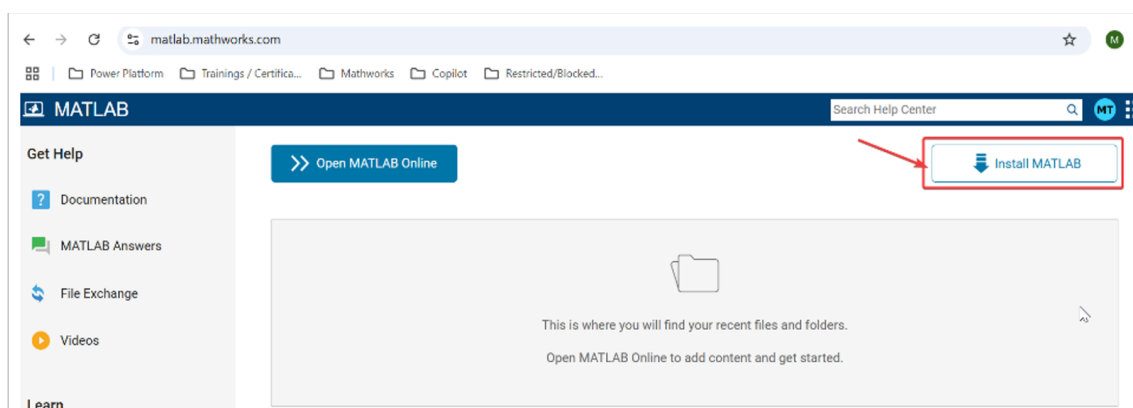
MathWorks login page

5.3 Download MATLAB

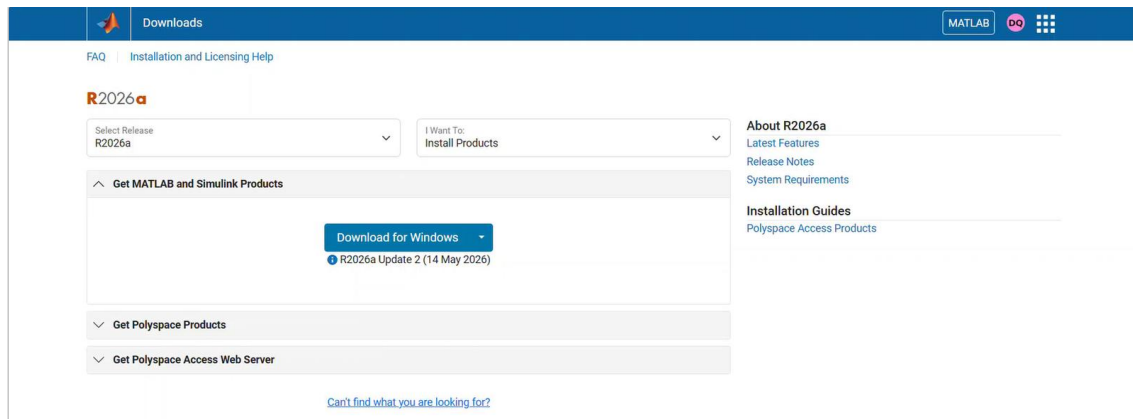
1. After logging in successfully, click **MATLAB** in the top navigation bar.
2. Click **Install MATLAB** to open the download page.
3. Download the installer for Windows or macOS. macOS participants must choose the Apple-silicon installer.



Click MATLAB



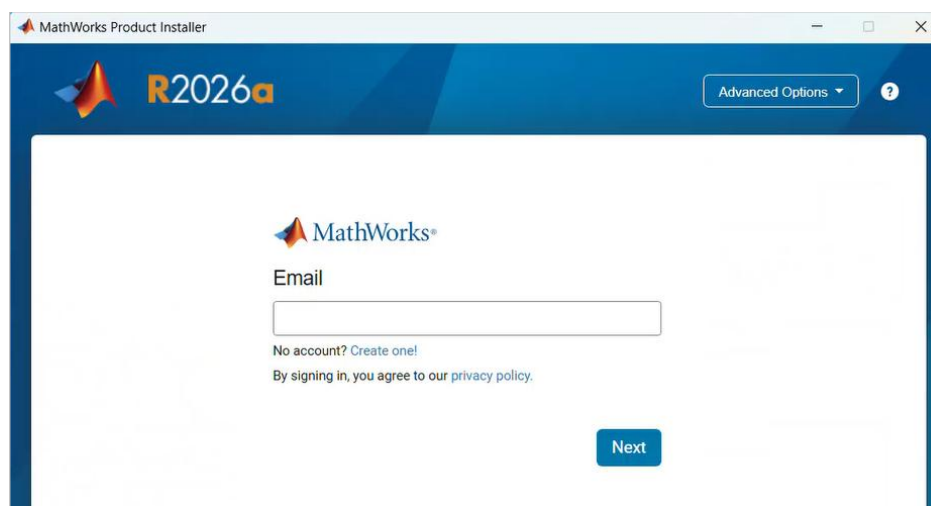
Install MATLAB



MATLAB download page

5.4 Run the MATLAB Installer

1. On Windows, double-click the installer. On macOS, open the downloaded DMG and launch InstallForMacOSAppleSilicon.
2. When prompted, sign in using your MathWorks credentials.
3. Select the trial license you want to activate.
4. Select the destination folder.
5. Choose Select All to install all products, or select at least the products listed in Section 3. The product list is the same on Windows and macOS.
6. Continue through the installer until installation is complete.



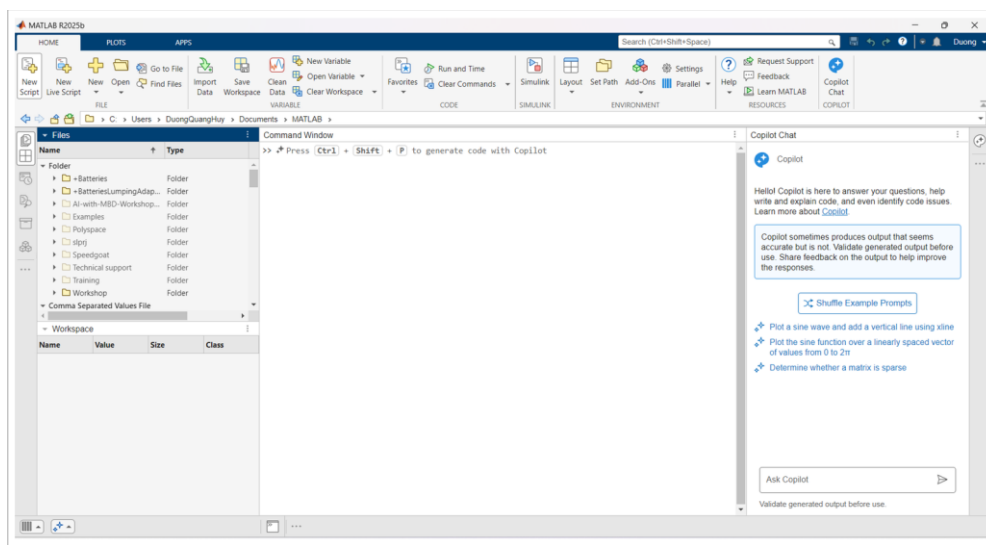
MATLAB Installer login prompt

License number selection

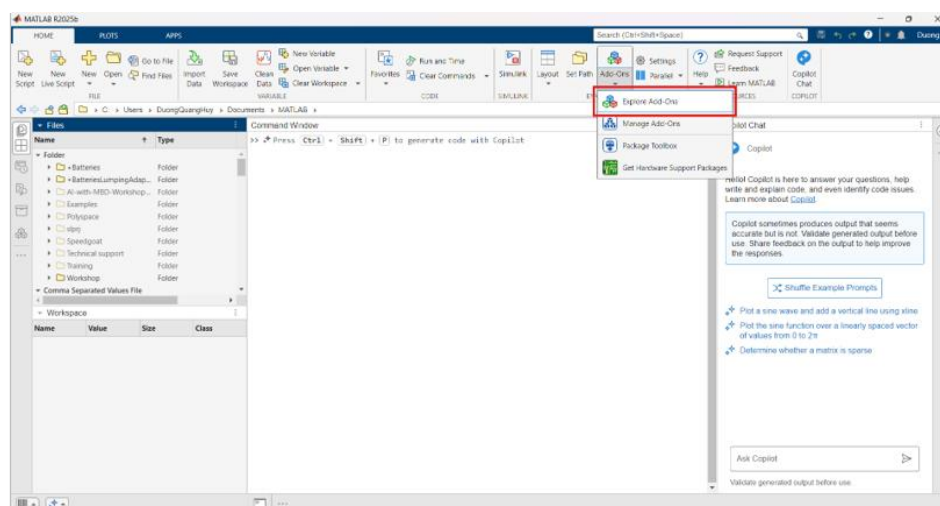


1. Open MATLAB.

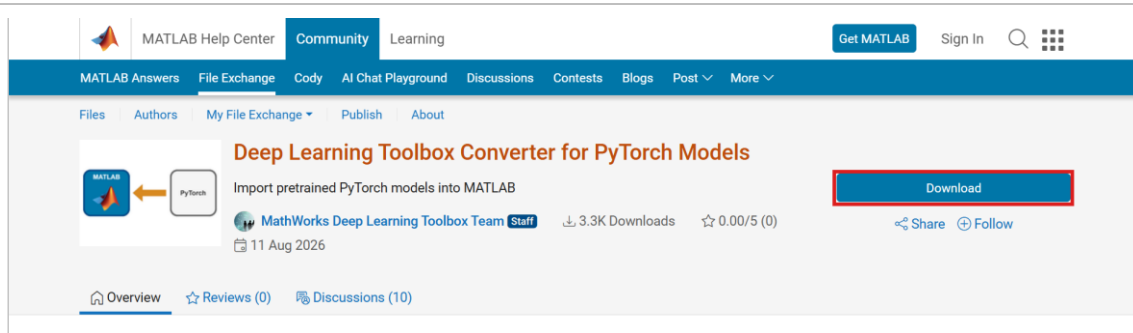
2. On the **Home** tab, click **Add-Ons** and then **Get Add-Ons** to open the Add-On Explorer.
3. Search for **Deep Learning Toolbox Converter for PyTorch Models**.
4. Click **Install** and follow the instructions.
5. Repeat the same process for:
 - Deep Learning Toolbox Model Compression Library
 - MATLAB Coder Interface for Deep Learning
 - MATLAB Support for MinGW-w64 C/C++/Fortran Compiler (Windows only)
 - macOS participants do not install MinGW. Install Xcode 16 or Xcode 26 separately from Apple before configuring MEX.



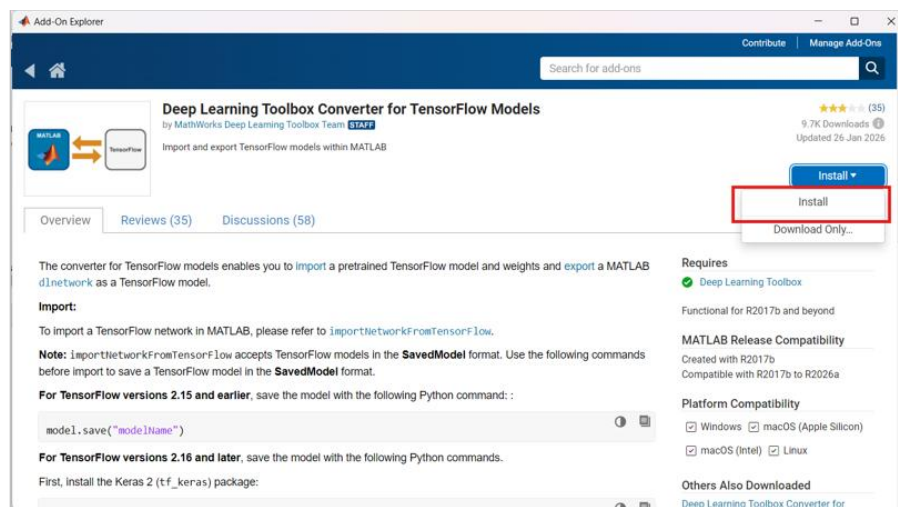
MATLAB desktop



Home tab > Add-Ons > Get Add-Ons



Deep Learning Toolbox Converter for PyTorch Models



Click Install on the add-on page (example add-on shown; the steps are identical)

Repeat the same process for the remaining MATLAB add-ons listed in Section 4. MinGW-w64 applies only to Windows; Xcode is installed separately on macOS.

5.6 Configure the C and C++ Compilers

1. In the MATLAB Command Window, configure both C and C++ compilers:

```
mex -setup C
mex -setup C++
```

2. Windows should report MinGW64 or a supported Microsoft compiler. macOS should report Apple Clang from Xcode 16 or Xcode 26.
3. If no supported compiler is found, Windows participants should install MinGW-w64 from Section 5.5. macOS participants should complete the Xcode steps below. A missing compiler does not block Parts 1 and 2, but desktop MEX validation in Part 3 will use instructor-prepared evidence.

5.6.1 Configure Xcode on macOS

1. Install the complete Xcode 16 or Xcode 26 application from Apple.
2. Open Xcode once and accept the license and any requested components.

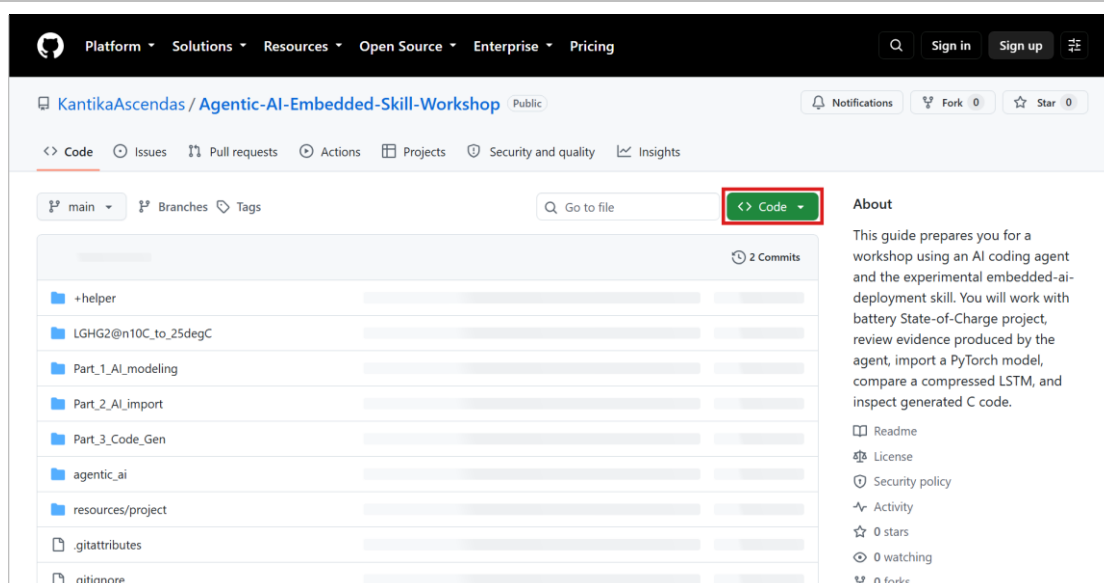
3. In Terminal, run `xcode-select -p`. The result should point to the active Xcode developer directory.
4. Return to MATLAB and run `mex -setup C` followed by `mex -setup C++`.
5. Confirm the selected compiler with `mex.getCompilerConfigurations("C++","Selected")`.

5.7 Download the Workshop Project

1. Open the GitHub repository:
<https://github.com/KantikaAscendas/Agentic-AI-Embedded-Skill-Workshop>
2. Click **Code** and choose **Download ZIP**, or clone the repository with Git:

```
git clone https://github.com/KantikaAscendas/Agentic-AI-Embedded-Skill-Workshop.git
```

3. Extract or clone the project to a short, writable local path. Windows example: `C:\Workshop\Agentic-AI-Embedded-Skill-Workshop`. macOS example: `~/Workshop/Agentic-AI-Embedded-Skill-Workshop`.
4. Do not work inside the ZIP file, on a network drive, or in a cloud-synchronised folder such as OneDrive or iCloud Drive. Code generation creates many files and can run slowly there.



GitHub workshop repository: Code > Download ZIP

★ Note

Download the repository **before** the event. The data files are large (about 25 MB) and venue Wi-Fi may be slow.

5.8 Verify the Setup

1. In MATLAB, open **Aimbdworkshop.prj** from the project folder. The project sets up the MATLAB path automatically.
2. Run the preflight script:

```
run("agentic_ai/scripts/workshopPreflight.m")
```

3. Confirm that the preflight reports `READY=true`. If only the host compiler is missing, continue as `READY WITH FALLBACK` and plan to use the instructor-prepared MEX evidence.

4. If a required MATLAB product or add-on reports MISSING, install it, restart MATLAB, and run the preflight again. A missing external Python installation is acceptable because prepared PyTorch reference data is included.
5. If the result is still READY=false for anything other than the compiler, take a screenshot and send it to the workshop coordinator before the event.

★ Next step

The preflight also checks that the embedded-ai-deployment Agent Skill is installed. This line stays MISSING until the AI coding agent is set up. Continue with Section 6 to install one supported AI coding agent; the MATLAB Agentic Toolkit and the skill itself are installed from inside the selected agent, following the “Workshop Instruction Embedded AI” participant guide.

6. Choose One AI Coding Agent

Install only one AI coding agent before the event. Codex is the workshop-validated default. Claude Code is a supported alternative. Google Antigravity CLI is an optional advanced alternative that may require manual MATLAB MCP configuration.

6.1 Option A ChatGPT Desktop with Codex

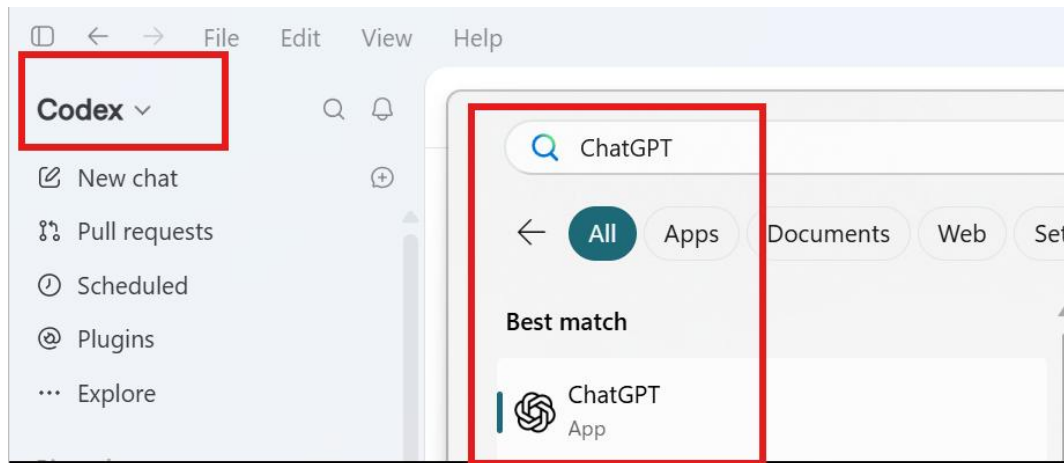
Install the ChatGPT desktop app for your operating system before the event. Use the official download page. Windows participants may also use Microsoft Store or Windows Package Manager when company policy permits it.

1. Go to <https://chatgpt.com/download> and select the Windows or macOS download. On Windows, you may alternatively run this command in PowerShell:

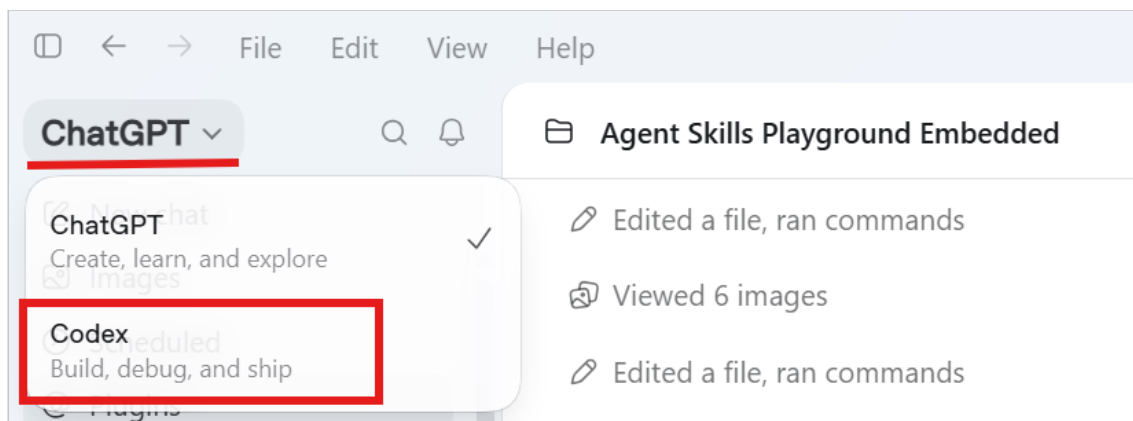
```
winget install --id 9PLM9XGG6VKS -s msstore
```

2. Run the installer, open ChatGPT from the Windows Start menu or macOS Applications folder, and sign in with your ChatGPT account.
3. Click the **ChatGPT** drop-down at the top-left of the app and select **Codex** (Build, debug, and ship). The label at the top-left changes to **Codex**.

Official ChatGPT desktop app for Windows and macOS: chatgpt.com/download



Open the ChatGPT app from the Start menu



Switch from ChatGPT to Codex in the top-left drop-down

6.2 Option A Codex CLI Fallback on Windows

The PowerShell installer below is a Windows fallback. On macOS, use the ChatGPT desktop app for this workshop unless your IT team has approved a separate Codex CLI installation.

1. Open PowerShell and run the official OpenAI installer:

```
powershell -ExecutionPolicy Bypass -c "irm https://chatgpt.com/codex/install.ps1 | iex"
```

2. Start Codex from a new PowerShell window:

```
codex
```

3. At first launch, select **Sign in with ChatGPT** and complete authentication.

★ Note

ExecutionPolicy Bypass applies only to that PowerShell process; it does not override company security controls. If installation is blocked, pair with another participant or follow along with the instructor.

6.3 Option B Claude Code CLI

Claude Code is supported by the MATLAB Agentic Toolkit. Install it only when you prefer Claude Code instead of Codex.

Windows PowerShell:

```
irm https://claude.ai/install.ps1 | iex
```

macOS Terminal:

```
curl -fsSL https://claude.ai/install.sh | bash
```

Verify with `claude --version`, open the workshop repository in Terminal or PowerShell, and run `claude`. Sign in when prompted. After the MATLAB Agentic Toolkit is installed, use `/skills` to confirm the required skills are visible.

6.4 Option C Google Antigravity CLI

Antigravity CLI supports MCP and Agent Skills, but it is not the validated workshop baseline. Manual MATLAB MCP and skill configuration may be required. Use this option only if you can complete the workshop preflight before the event.

Windows PowerShell:

```
irm https://antigravity.google/cli/install.ps1 | iex
```

macOS Terminal:

```
curl -fsSL https://antigravity.google/cli/install.sh | bash
```

Start with `agy`. Use `/mcp` to verify the MATLAB MCP server and `/skills` to verify the embedded AI skill. Repository-local skills belong under `.agents/skills`. If either check fails, use Codex or Claude Code during the workshop.

6.5 Account and Usage Guidance

Participants do not need all three agents. Use an existing subscription when available. A free tier may be sufficient for the short gated exercises, but usage limits differ by provider and can change. If an allowance is reached, continue with the instructor-prepared scripts and results. Never share account credentials or API keys.

7. Pre-Workshop Checklist

Tick each item once it is done:

- ☐ MATLAB R2026a installed on supported Windows or an Apple-silicon Mac, with all products listed in Section 3
- ☐ The three AI/code-generation add-ons installed; Windows participants also installed MinGW-w64 when needed
- ☐ `mex -setup C` and `mex -setup C++` report a configured compiler, or `READY WITH FALLBACK` is understood
- ☐ Workshop repository downloaded to a short local path
- ☐ `Aimbdworkshop.prj` opens without errors

- ☐ workshopPreflight.m reports every MATLAB product and add-on as OK
- ☐ One AI coding agent installed and opens after sign-in: Codex, Claude Code, or Antigravity CLI
- ☐ Laptop charger packed; MathWorks and selected AI-agent account sign-in details known